UNIT I INTRODUCTION TO BIG DATA

Big Data – Definition, Characteristic Features – Big Data Applications - Big Data vs Traditional Data - Risks of Big Data - Structure of Big Data - Challenges of Conventional Systems - Web Data – Evolution of Analytic Scalability - Evolution of Analytic Processes, Tools and methods - Analysis vs Reporting - Modern Data Analytic Tools.

Definition

1. "Big Data is a collection of large datasets that cannot be adequately processed using traditional processing techniques".

2. "Big data refers to data sets whose size is beyond the ability of typical database software tools to capture, store, manage and analyze".

Examples Of Big Data

The **New York Stock Exchange** generates about ***one terabyte*** of new trade data per day.

• **Social Media Impact**

Statistics shows that ***500+terabytes*** of new data gets into the databases of social media site **Facebook**, every day. This data is mainly generated in terms of photo and video uploads, message exchanges, placing comments etc.

Categories Of 'Big Data'

1. **Structured**
2. **Unstructured**
3. **Semi-structured**

Structured

Any data that can be stored, accessed and processed in the form of fixed format is termed as 'structured' data.

Example:

An 'Employee' table in a database is an example of Structured Data

Employee_ID	Employee_Name	Gender	Department	Salary_In_lacs
2365	Rajesh Kulkarni	Male	Finance	650000
3398	Pratibha Joshi	Female	Admin	650000

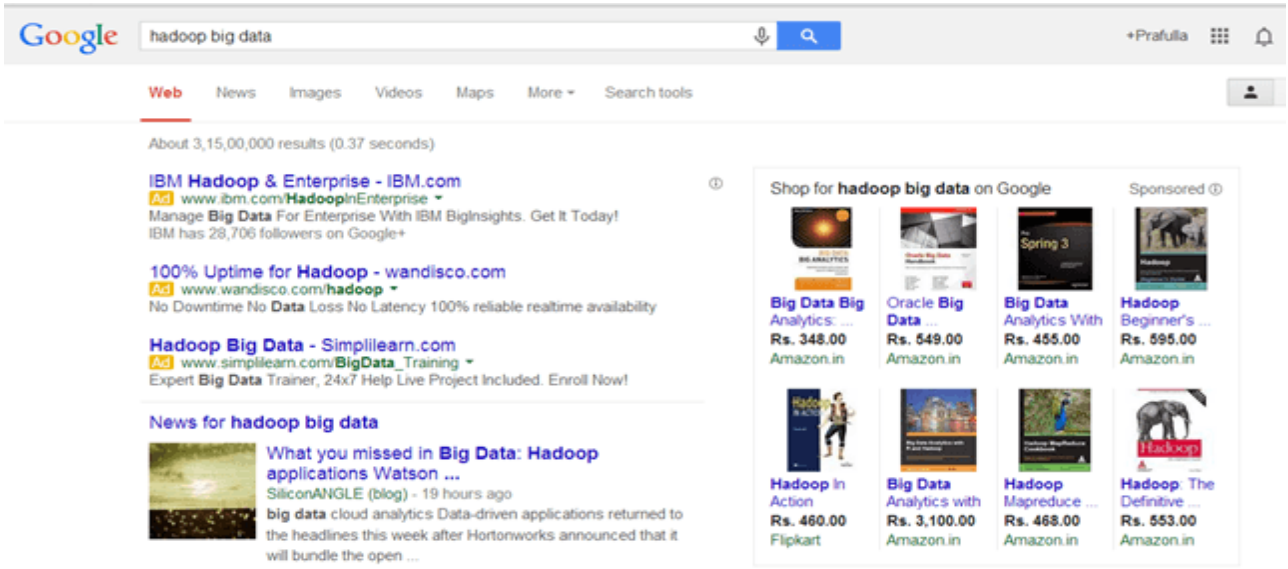
7465	Shushil Roy	Male	Admin	500000
7500	Shubhojit Das	Male	Finance	500000
7699	Priya Sane	Female	Finance	550000

Unstructured

Any data with unknown form or the structure is classified as unstructured data.

Examples Of Un-structured Data

Output returned by 'Google Search'



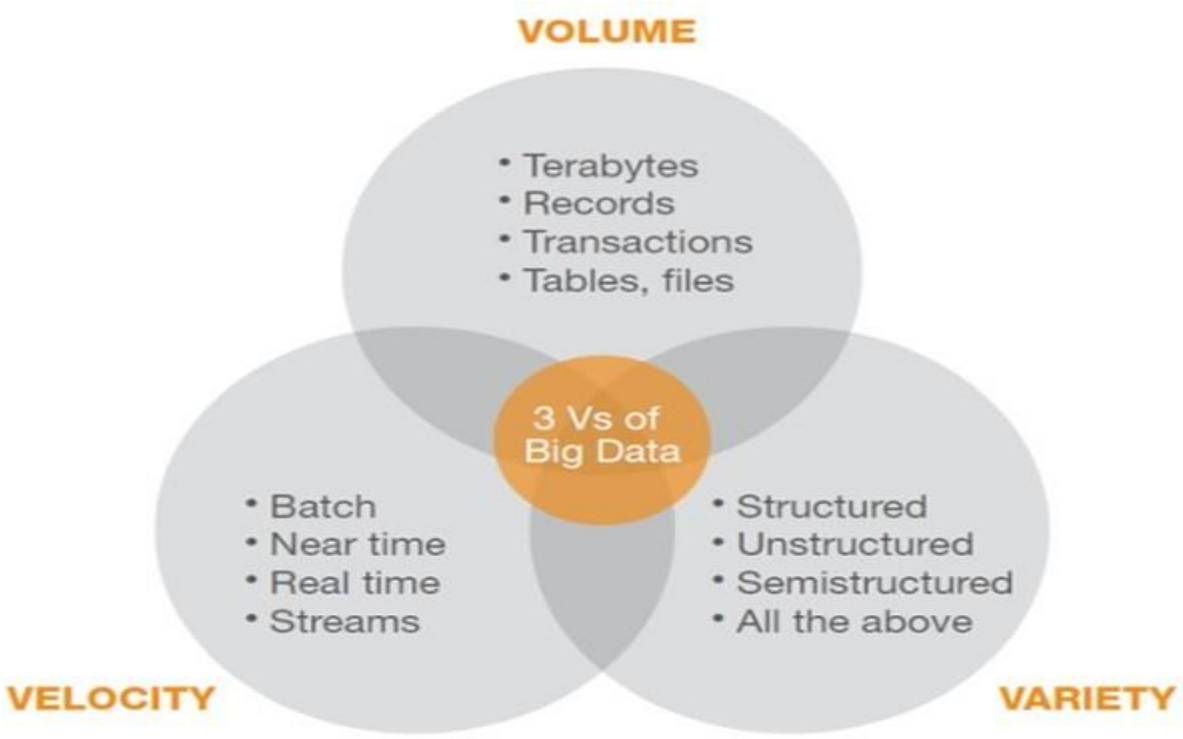
Semi-structured

Semi-structured data can contain both the forms of data. Example of semi-structured data is a data represented in XML file.

Personal data stored in a XML file-

```
<rec><name>Prashant Rao</name><sex>Male</sex><age>35</age></rec>
<rec><name>Seema R.</name><sex>Female</sex><age>41</age></rec>
<rec><name>Satish Mane</name><sex>Male</sex><age>29</age></rec>
<rec><name>Subrato Roy</name><sex>Male</sex><age>26</age></rec>
<rec><name>Jeremiah J.</name><sex>Male</sex><age>35</age></rec>
```

Characteristics of Big Data



i)Volume

- The name 'Big Data' itself is related to size which is enormous.
- Size of data plays vital role in determining value out of data.
- Also, whether a particular data can actually be considered as a Big Data or not, is dependent upon volume of data.
- Hence, **'Volume'** is one characteristic which needs to be considered while dealing with 'Big Data'.

(ii)Variety

- Variety refers to heterogeneous sources and the nature of data, both structured and unstructured.
- During earlier days, spreadsheets and databases were the only sources of data considered by most of the applications.
- Nowadays, data in the form of emails, photos, videos, monitoring devices, PDFs, audio, etc. is also being considered in the analysis applications.

(iii)Velocity

- The term **'velocity'** refers to the speed of generation of data. How fast the data is generated and processed to meet the demands, determines real potential in the data.
- Big Data Velocity deals with the speed at which data flows in from sources like business processes, application logs, networks and social media sites, sensors, Mobile devices, etc.

Applications of Big Data

Banking

- Large amounts of data streaming in from countless sources, banks have to find out unique and innovative ways to manage big data.
- It's important to analyze customers needs and provide them service as per their requirements, and minimize risk and fraud while maintaining regulatory compliance.
- Big data deal with financial institutions to do one step from the advanced analytics.

Government

- When government agencies are controlling and applying analytics to their big data, they have improvised a lot in terms of managing utilities, running agencies, dealing with traffic congestion.

Education

- Big Data provides a significant impact on school systems, students and curriculums.
- By analyzing big data, they can identify risk at students, ensuring student's progress, and can implement an improved system for evaluation and support of teachers and principals in their teachings.

Health Care

- When it comes to health care in terms of Patient records, treatment plans, prescription information etc., everything needs to be done quickly and accurately.

- Effective management results in good health care to uncover hidden insights that improve patient care.

Manufacturing

- Manufacturers can improve their quality and output while minimizing waste where processes are known as the main key factors in today's highly competitive market.
- Several manufacturers are working on analytics where they can solve problems faster and make more agile business decisions.

Retail

- Customer relationship maintenance is the biggest challenge in the retail industry and the best way to manage will be to manage big data.
- Retailers must have unique marketing ideas to sell their products to customers, the most effective way to handle transactions, and applying innovative ideas using Big Data to improve their business.

Big Data Vs Traditional Data

There are some important ways that big data is different from traditional data sources.

- First, **big data is automatically generated by a machine. Instead of a person being involved in creating new data, it's generated purely by machines in an automated way**.
- In traditional data sources, there was always a person involved.

  Consider retail or bank transactions, telephone call detail records, product shipments, or invoice payments. All of those involve a person doing

something in order for a data record to be generated. Somebody had to deposit money, or make a purchase, or make a phone call, or send a shipment, or make a payment. In each case, there is a person who is taking action as part of the process of new data being created. This is not so for big data in many cases. A lot of sources of big data are generated without any human interaction at all.

- Second, **big data is an entirely new source of data. It is not simply an extended collection of existing data**. For example, with the use of the Internet, customers can now execute a transaction with a bank or retailer online. But the transactions they execute are not fundamentally different transactions from what they would have done traditionally. They've simply executed the transactions through a different channel.
- Third, **many big data sources are not designed to be friendly**. Take text streams from a social media site. There is no way to ask users to follow certain standards of grammar, or sentence ordering, or vocabulary. It can be difficult to work with such data at best and very, very ugly at worst.

Risks Of Big Data

1. Data Security

The bigger your data, the bigger the target it presents to criminals with the tools to steal and sell it. In the case of Target, hackers stole credit and debit card information of 40 million customers, as well as personal identifying information such as email and geographical addresses of up to 110 million people.

2. Data Privacy

Closely related to the issue of security is privacy. But in addition to ensuring that people's personal data are safe from criminals, we need to be sure that the sensitive

information that are storing and collecting isn't going to be misuse by yourself or by people to whom you have delegated responsibility for analyzing and reporting on it.

3. Costs

Data collection, aggregation, storage, analysis, and reporting all cost money.

Challenges of Conventional Systems

In the past, the term 'Analytics' has been used in the business intelligence world to provide tools and intelligence to gain insight into the data through fast, consistent, interactive access to a wide variety of possible views of information. The main challenge in the traditional approach is how to find all the hidden information through the vast amount of data.

- Traditional Analytics analyzes **the known data environment** that too the data that is well understood. It **cannot work on unstructured data** efficiently.
- Traditional Analytics is **built on top of the relational data model**, relationships between the subjects of interests have been created inside the system and the analysis is done based on them. This approach will not adequate for big data analytics.
- Traditional analytics is **batch oriented** and we need to wait for ETL (extract, transform and load) and transformation jobs to complete before the required insight is obtained.
- Parallelism in a traditional analytics system is achieved through costly **hardware like MPP** (Massive Parallel Processing) systems
- Insufficient support of aggregated summaries of data

Apart from these challenges others are categorized as

Data challenges

- Volume, velocity, veracity, variety
- Data discovery
- Scalability

Process challenges

- Capturing data
- Aligning data from different sources
- Transforming data into suitable form for data analysis
- Modeling data(mathematically, simulation)
- Understanding output, visualizing results and display issues on mobile devices

Management challenges

- Security
- Privacy
- Governance
- Ethical issues

Web Data

Web data is the data that comes from large or diverse number of sources.

Imagine the Possibilities

Imagine knowing everything customers do as they go through the process of doing business with the organization. Not just what they buy, but what they are thinking about buying along with what key decision criteria they use. Such **knowledge enables a new level of understanding about the customers and a new level of**

interaction with the customers. It allows you to meet their needs more quickly and keep them satisfied.

- Imagine you are a retailer. Imagine walking with customers and recording every place they go, every item they look at, every item they pick up, every item they put in the cart and then take back out. Imagine knowing whether they read nutritional information, if they look at laundry instructions, if they read the promotional brochure on the shelf, or if they look at other information made available to them in the store.
- Imagine you are a bank. Imagine being able to identify every credit card option customers considered. Imagine being able to understand if it was a reward program, interest rates, or annual fees that drove their choice. Imagine knowing what they say about each product after they own it.
- Imagine you are an airline. Imagine being able to identify every flight customers viewed before choosing their final itinerary. Imagine knowing if they cared more about price or convenience. Imagine knowing all the destinations they consider and when they first consider them.

A Fundamentally New Source of Information

The beauty of exploring customer's detailed web behavior is that it moves beyond just knowing what they buy. We can gain insight into how they made their decisions. Instead of seeing just the results, we have visibility into the entire buying process.

One of the **most exciting aspects of web data is that it provides factual information on customer preferences, future intentions, and motivations that are virtually impossible to get from other sources outside of a direct conversation or survey.**

What Data Should Be Collected?

Any action that a customer takes while interacting with an organization should be captured if it is possible to capture it. That means detailed event history from any customer touch point. Common touch points today include web sites, mobile apps, and social media.

WHAT WEB DATA REVEALS

There are a number of specific areas where **web data can help organizations understand their customers better than is possible without web data**. Without taming this source of big data, such insights will be very difficult.

Shopping Behaviors

A good starting point to understand shopping behavior is **identifying how customers come to a site to begin shopping. What search engine do they use? What specific search terms are entered? Do they use a bookmark they created previously? Analytic professionals can take this information and look for patterns in terms of which search terms, search engines, and referring sites are associated with higher sales rates.**

Customer Purchase Paths and Preferences

Using web data, **it is possible to identify the ways customers arrive at their buying decisions by watching how they navigate a site. It is also possible to gain insight into their preferences.**

Consider for a moment an airline. An airline can tell a number of things about preferences based on the ticket that is booked. For example, how far in advance was the ticket booked? What fare class was booked? Did the trip span a weekend or not? This is all useful, but an airline can get even more from web data.

Feedback Behaviors

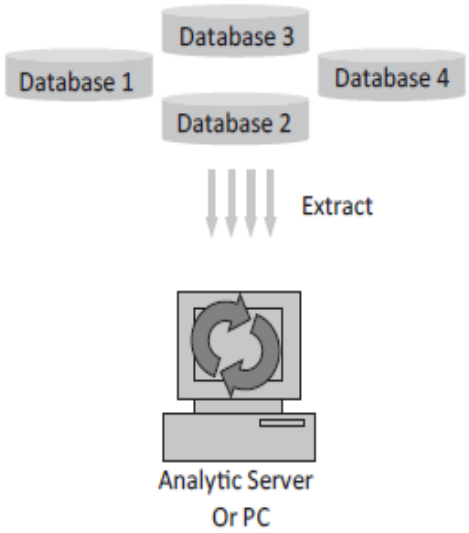
Some of the best information customers can provide is **detailed feedback on products and services**. Simply the fact that customers are willing to take the time to do so indicates that they are engaged with a brand. By using text mining to understand the tone, intent, and topic of a customer's feedback, a better picture of what is important to that individual begins to form.

Evolution of Analytic Scalability

The Convergence of the Analytic and Data Environment

1. Traditional Analytic Architecture

Traditional analytics collects data from heterogeneous data sources and we had to pull all data together into a separate analytics environment to do analysis which can be an analytical server or a personal computer with more computing capability. The heavy processing occurs in the analytic environment as shown in figure.



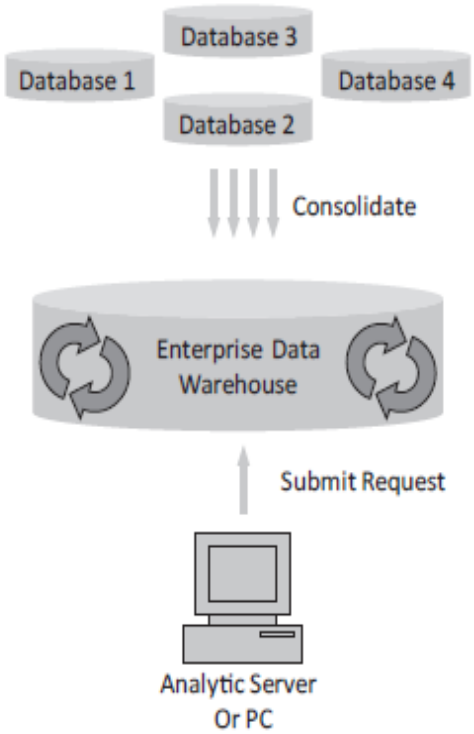
In traditional architectures, the heavy processing occurs in the analytic environment. This may even be a PC!

In such environments, **shipping of data becomes must, which might result in issues related with security of data and its confidentiality**.

2. Modern In-Database Architecture

Data from heterogeneous sources are collected, transformed and loaded into data warehouse for final analysis by decision makers. The processing stays in the database where the data has been consolidated.

The data is presented in aggregated form for querying. Queries from users are submitted to OLAP (online analytical processing) engines for execution. Moreover the data in consolidated form are free from anomalies, since they are preprocessed before loading into warehouses which may be used directly for analysis. The modern in-database architecture is shown in figure.



In an in-database environment, the processing stays in the database where the data has been consolidated. The user's machine just submits the request; it doesn't do heavy lifting.

3. Massive Parallel Processing (MPP)

Massive Parallel Processing (MPP) is the "**shared nothing**" approach of parallel computing. In this type of computing, the process is being done by many CPUs working in parallel to execute a single program. One of the most significant differences between **Symmetric Multi-Processing or SMP and Massive Parallel Processing** is that with MPP, each of the many CPUs has its own memory to assist it in preventing a possible hold up that the user may experience with using SMP when all of the CPUs attempt to access the memory simultaneously. The MPP architecture is given in figure.

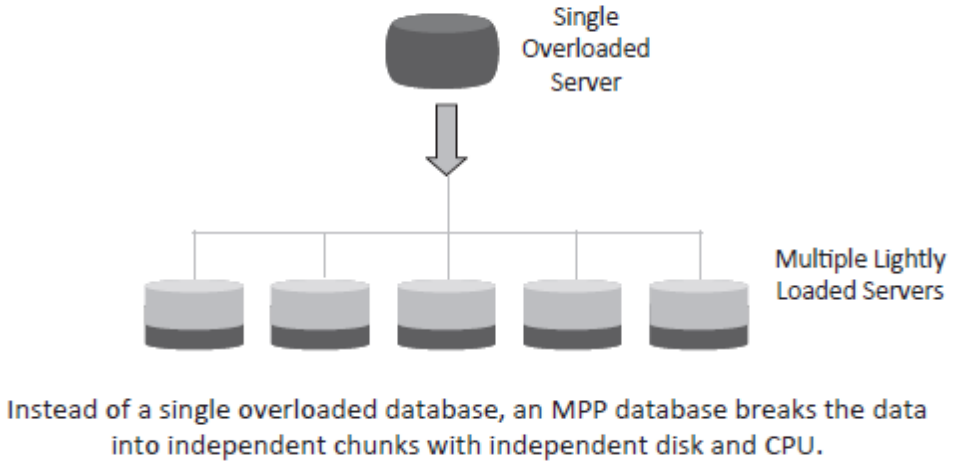


Fig: Massively Parallel Processing System Data Storage

The salient feature of MPP systems are:

• Loosely coupled nodes

• Nodes linked together by a high speed connection

• Each node has its own memory

• Disks are not shared, each being attached to only one node – shared nothing architectures

4. The Cloud Computing

- Cloud computing is the delivery of computing services over the Internet.
- Cloud services allow individuals and businesses to use software and hardware that are managed by third parties at remote locations.
- Examples of cloud services include online file storage, social networking sites, webmail, and online business applications.
- The cloud computing model allows access to information and computer resources from anywhere that a network connection is available.
- Cloud computing provides a shared pool of resources, including data storage space, networks, computer processing power, and specialized corporate and user applications.

The five essential characteristics of a cloud environment specified by NIST are:

1. **On-demand self-service**

2. Broad network access

3. Resource pooling

4. Rapid elasticity

5. Measured service

There are two types of cloud environment:

1. Public Cloud

– The services and infrastructure are provided off-site over the internet

– Greatest level of efficiency in shared resources

– Less secured and more vulnerable than private clouds

2. Private Cloud

– Infrastructure operated solely for a single organization

– The same features of a public cloud

- Offer the greatest level of security and control

– Necessary to purchase and own the entire cloud infrastructure

Categories of Cloud Computing

- Cloud computing delivers **infrastructure, platform, and software (application) as services**, which are made available as subscription-based services in a pay-as-you-go model to consumers.

- **IaaS gives users automated and scalable environments with extreme flexibility and control**, while **PaaS provides a framework for quickly developing and deploying applications** by automating infrastructure provisioning and management. Eliminating the need to install and run programs on individual devices, **SaaS makes applications available through the internet.**

5. Grid Computing

- Grid Computing can be defined as a network of computers working together to perform a task that would rather be difficult for a single machine.
- All machines on that network work under the same protocol to act like a virtual supercomputer. The task that they work on may include analyzing huge datasets or simulating situations which require high computing power.
- It is a form of Parallel Computing where instead of many CPU cores on a single machine, it contains multiple cores spread across various locations.

6. Hadoop

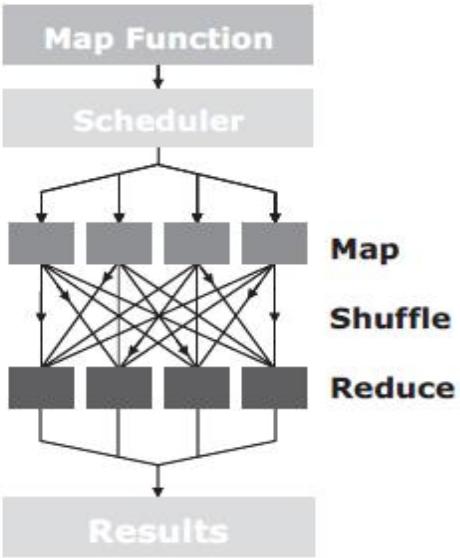
Apache Hadoop is an open-source software framework for storage and large-scale processing of data-sets on clusters of commodity hardware.

Two main building blocks inside this runtime environment are **MapReduce and Hadoop Distributed File System (HDFS).**

Hadoop MapReduce is a software framework for easily writing applications which process vast amounts of data (multi-terabyte data-sets) in-parallel on large clusters (thousands of nodes) of commodity hardware in a reliable, fault-tolerant manner.

A MapReduce job usually splits the input data-set into independent chunks which are processed by the map tasks in a completely parallel manner. The framework sorts the outputs of the maps, which are then input to the reduce tasks.

Typically both the input and the output of the job are stored in a file-system. The framework takes care of scheduling tasks, monitoring them and re-executes the failed tasks.



MapReduce breaks a job into many pieces and runs them independently.

Evolution of Analytic Processes, Tools and Methods

Evolution of Analytic Processes

- Upgrading technologies won't provide a lot of value, if the same old analytical processes remain in place
  1. Change the process of configuring and maintaining workspace

     The Analytic SandBox
  2. Consistently leverage a database platform through a sandbox

     Enterprise Analytic Data Set (EADS)
  3. Necessary to keep scores up to date on a daily

     Embedded Scoring

Analytic Sandbox Definition

An **analytic sandbox provides a set of resources with which analysis can be done to answer critical business questions**.

An analytic sandbox is ideal for

- data exploration
- development of analytical processes
- Proof of concepts, and prototyping.

Analytic Sandbox Benefits

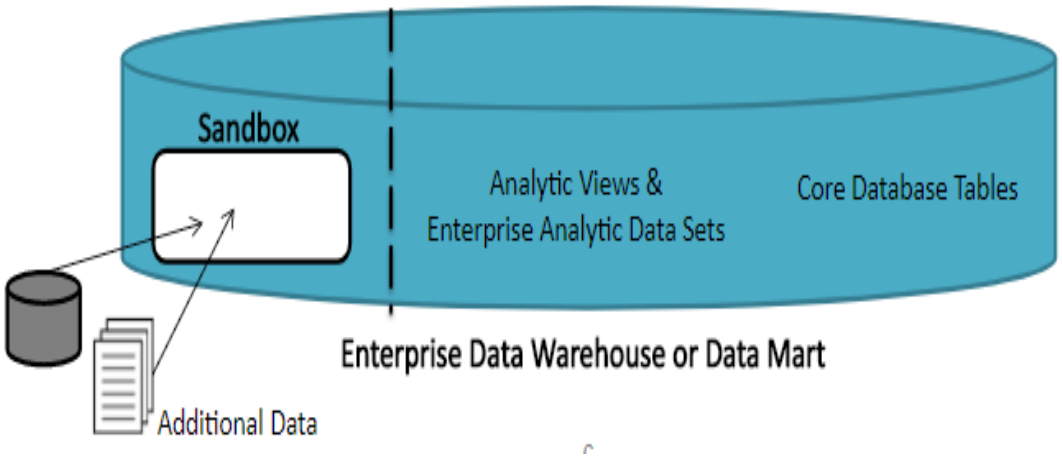
Independence. Analytic professionals will be able to work independently on the database system without needing to continually go back and ask for permissions for specific projects.

Flexibility. Analytic professionals will have the flexibility to use whatever business intelligence, statistical analysis, or visualization tools that they need to use.

Efficiency. Analytic professionals will be able to control the existing enterprise data warehouse or data mart, without having to move or migrate data.

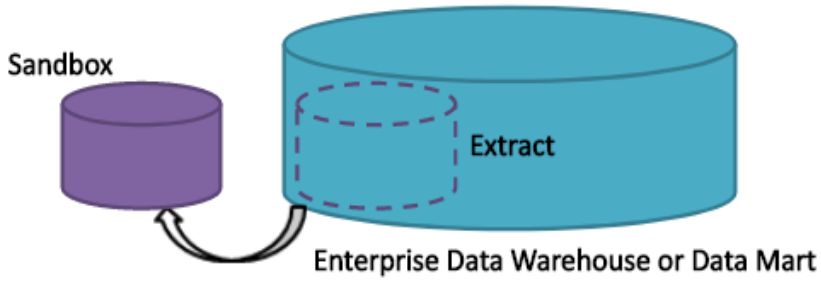
An Internal Sandbox

- A portion of an enterprise data warehouse or data mart is carved out to serve as the analytic sandbox
  - Strength
    - Leverage existing hardware resources and infrastructure already in place
    - Ability to directly join production data with sandbox data
    - Cost-effective since no new hardware is needed
  - Weaknesses
    - An additional load on the existing enterprise data warehouse or data mart
    - Can be constrained by production policies and procedures



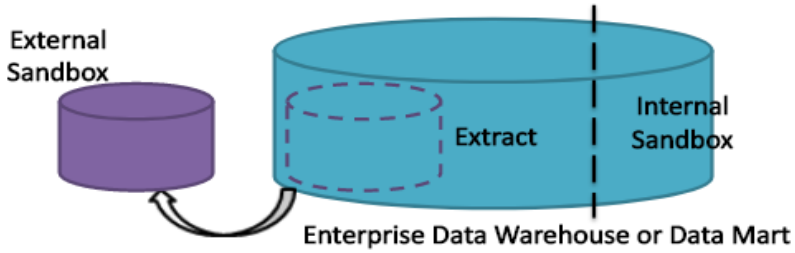
An External Sandbox

- A physically separate analytic sandbox is created for testing and development of analytic processes
  - Strength
    - A stand-alone environment, no impact on other processes
    - Reduce workload management
  - Weaknesses
    - The additional cost of the stand-alone system
    - Some data movement



A Hybrid Sandbox

- The combination of an internal sandbox and an external sandbox
  - Strength
    - Flexibility in the approach taken for an analysis
    - Can be run in a 'pseudo-production' mode temporarily
  - Weaknesses
    - Maintain both an internal and external sandbox environment
    - Two-way data feeds may be required, which adds complexity



Analytic Data Set

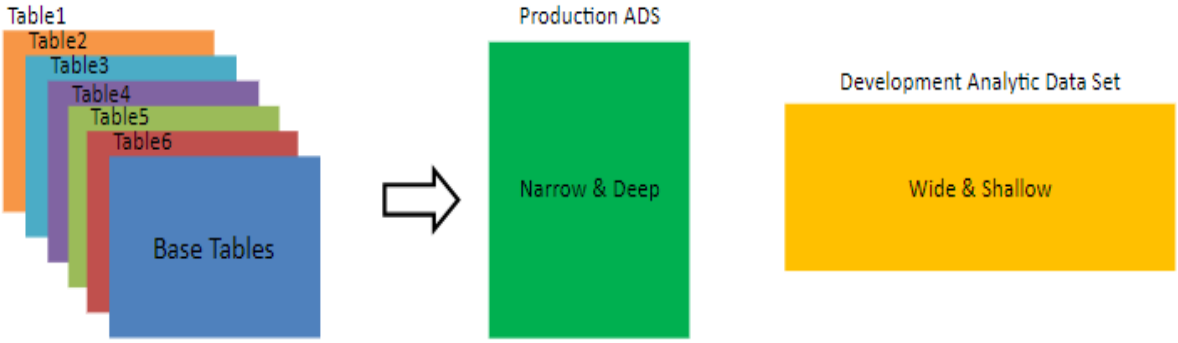
Definition

- The data that is pulled together in order to create an analysis or model
  - In the format required for the specific analysis at hand
  - Generated by transforming, aggregating, and combining data
  - Help to bridge the gap between efficient storage and ease of use

Two Types of Analytic Data Sets

- A development ADS
  - Used to build an analytic process
  - Have many variables or metrics within it
  - Very wide but not very deep

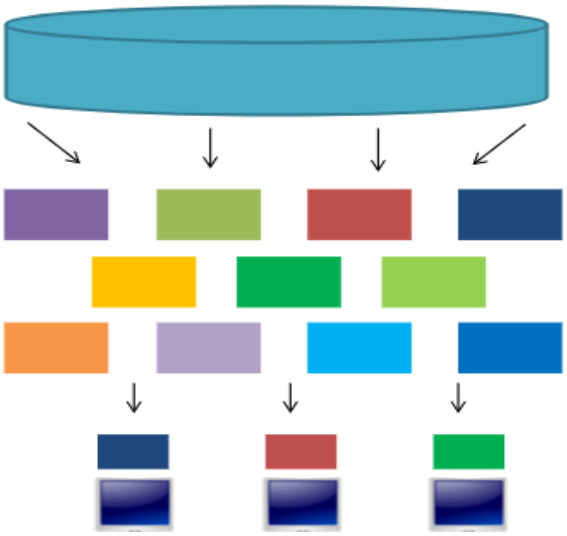
- Production analysis data set
  - Needed for scoring and deployment
  - Contain only the specific metrics that were actually in the final solution
  - Not very wide but very deep



Traditional Analytic Data Sets

- All analytic data sets are created outside of the database
  - Each analytic professional creates their own data sets independently
  - The risk of inconsistencies
  - The repetitious work

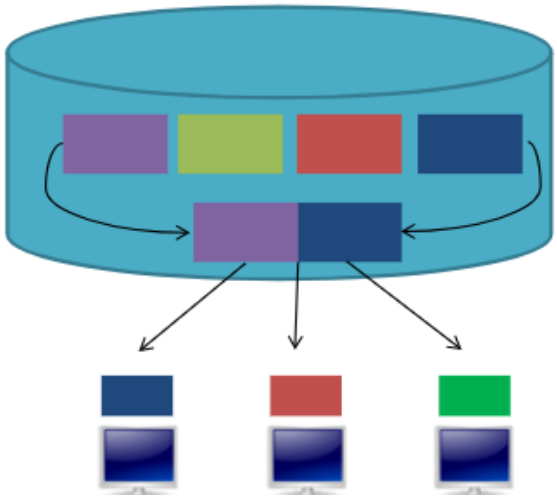
A dedicated ADS is generated outside the database for every project



Enterprise Analytic Data Sets

- A shared and reusable set of centralized, standardized analytic data sets for use in analytics
  - A standardized view of data to support multiple analysis efforts
  - Streamline the data preparation process
  - Provide grate consistency, accuracy, and visibility to analytics processes
  - Build once, use many

Centralized ADS tables and views are utilized across many projects



EADS Logical View:

Customer ADS Table

Customer	Total Sales	Total Purchases	Home-owners	Gender	Mail Responder	E-mail Opt in

EADS Potential Physical View:

Customer Sales

Customer	Total Sales	Total Purchases

Customer Demographics

Customer	Home-owner	Gender

Customer Sales

Customer	Mail Responder	E-mail Opt in

It could very well be stored differently!

For updating an EADS

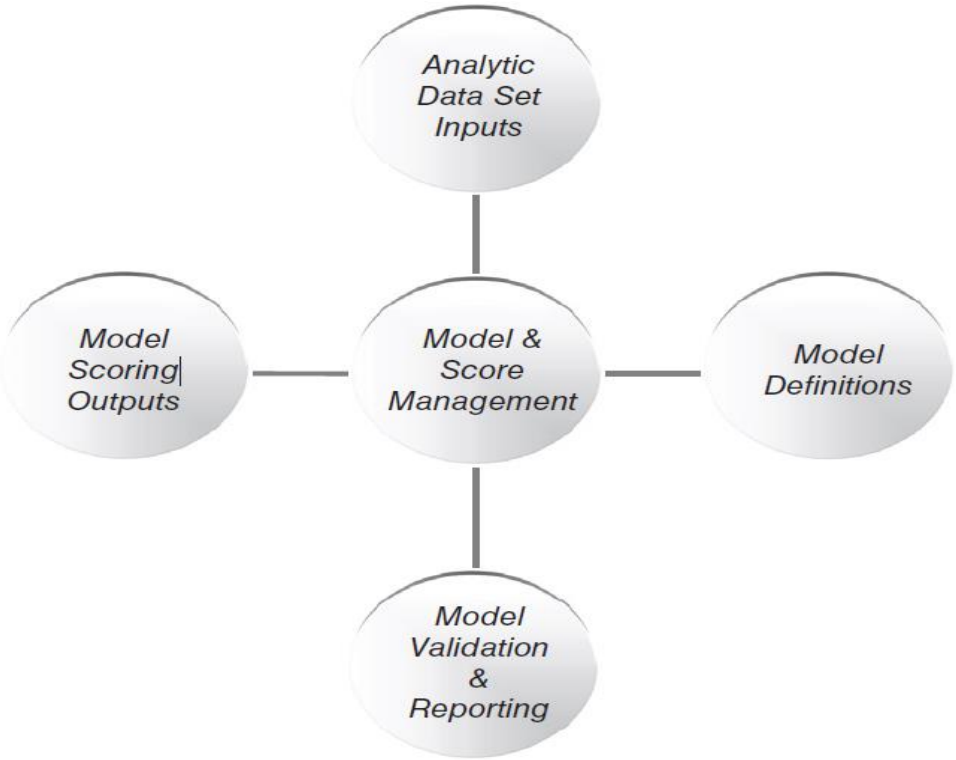
Scoring

After model creation based on historical data, it can be applied to new data in order to make predictions about unseen behavior.

The process of using a model to make predictions about behavior that has yet to happen is called "scoring."

The output of the model, the prediction, is called a score. Scores can take just about any form, from numbers to strings to entire data structures, but the most common scores are numbers.

There are **four primary components required to effectively manage all of the analytic processes an enterprise develops**. The components include **analytic data set inputs, model definitions, model validation and reporting, and model scoring output**. There are commercially available tools to help with model and score management, or a custom solution can be built to address an organization's specific needs.



Evolution of Analytic Tools and Methods

Evolution of Analytic Methods

Many of the commonly used analytical and modeling approaches have been in use for many years. Some, such as **linear regression or decision trees, are effective and relevant, but relatively simple to implement.**

Until the advent of computers, it wasn't feasible to run many iterations of a model or to try highly advanced methods. Today it is possible to run many iterations of many algorithms against large and robust data sets.

Often, analytic professionals will simply do a lot more of the same long-standing methods as a result of today's new scalability.

While methods will continue to evolve, these include **ensemble methods, commodity modeling, and analysis of text data.**

Ensemble Methods

Ensemble approaches are straightforward conceptually.

Instead of building a single model with a single technique, multiple models are built using multiple techniques. Once the results from all of the models are known, all of the results are combined together to come up with a final answer.

The process of combining the various results can be anything from a simple average of each model's predictions to a much more complex formula. It is important to note that **ensemble models go beyond picking the best individual performer from a set of models. They actually combine the results of multiple models in order to get to a single final answer.**

The power of ensemble models stems from the fact that different techniques have different strengths and weaknesses. Certain types of customers, for example, may be scored poorly by one technique but very well by another.

For example, perhaps linear regression, logistic regression, a decision tree, and a neural network are all created to predict the likelihood of a customer purchasing a given product. The scores from each model would be combined into a final score in an ensemble approach. Often, that combination can beat the individual models at predicting the purchases.

Commodity Models

Commodity models might be done via a simple stepwise analysis procedure on a mostly automated basis. The **goal of a commodity model is not to get the best model, but to quickly get a model that will lead to a better result** than if there had been no model at all.

Commodity models can be quite useful and can extend the impact of analytics within an organization. Traditionally, building models was a time-intensive task.

As a result, it was expensive. Analysts would spend weeks or months just getting data together and then more time running models against the data.

Analytic professionals are **making use of a modern environment including a scalable sandbox, as well as modern processes including enterprise analytic datasets.**

Text Analysis

One of the most rapidly growing methods utilized by organizations today is the analysis of text and other unstructured data sources. A lot of big data falls into these classifications. **Text analysis, as the name implies, takes some sort of text as input**. This text can be written material like an e-mail, transcribed material such as a medical dictation, or even text that has been scanned from a hard copy and converted to electric form like old courthouse records.

The reason text analysis has grown in prominence is because of the wealth of new sources of text data. In recent years, everything from e-mails, to social media commentary from sites like Facebook and Twitter, to online inquiries, to text messages, to call center conversations is captured in bulk.

Making sense of all this text is no easy task. **There are issues to address when parsing it, issues to address when identifying context, and issues with defining meaningful patterns.**

Organizations are starting to have more text and unstructured data available to them than they do traditional, structured data. This type of data cannot be ignored. Text is a very common type of big data and text analysis tools and methods have come a long way. Today there are tools that will help you parse text into its component words and phrases and then assist in determining the meaning of those words and phrases. Popular commercial text analysis tools include those offered by **Attensity, Clarabridge, SAS, and SPSS**.

Once the text is parsed into its components, there are methods that will help identify the sentiment or meaning of those components and find trends within them. It is also common to take parsed text and feed summary statistics about it into other models. For example, how many of a given customer's e-mails were positive or negative in tone? How often did a given customer focus on a specific product line in his or her communications? This is essentially creating some structured information out of the raw, unstructured data. **This process of parsing and structuring text is often called information extraction.**

The Evolution of Analytic Tools

There are now **tools to manage deployment of analysis, to manage and administer the analytic servers and software that analytic professionals utilize, and to convert code from one language to another.**

There are also a number of commercial analytics packages available today. While the market leaders are SAS and SPSS, many other analytic software tools are also available.

In addition, there are now open source analytic tools as well.

Categories of Data Analytics Tools

There are basically 2 types of Data Analytics Tools:

a. Statistical data analysis Tool

All commercial Data Analytics tools come with graphical user interfaces. With the help of the evolved tools, the focus has shifted from coding to utilities. With the use of packages like point solutions, tasks can be accomplished very easily.

GUI is robust, bug free, optimized and allows analytic process development at a speed that equals or exceeds hard coding. Real analytic professionals do whatever is best to get a job done as accurately and efficiently as possible. Tools can help analytic professionals be more efficient while freeing up time to focus on analysis methods instead of writing code.

One big risk with user interfaces overlaps with one of their key strengths. It is easy to generate code in user interface; however, the ability to generate code quickly also makes it easy to generate bad code quickly. If a user is not proficient, he or she can accidentally create code through a user interface that is doing something totally different from what was intended.

b. Data Visualization Tool

The results obtained from the analysis of the data need to represented in forms that are useful for the user**. Visualization tools enable professionals to create an interactive, visual analytics**. An analytic professional will routinely need to explain complex analytical results to non-technical business people. Anything that can help this to be done more effectively is a good thing. Data visualization falls into this category. Many people would rather see a visual depiction of a decision tree model than a long list of business rules. This is where visualization helps.

The History of Open Source

Open-source software packages have been around for some time. **Open source packages are available to the public for free via download. In addition, the actual source code is also available so that users can customize and add on to the software as desired.** There are examples of widely adopted and highly successful open source applications. Firefox, the web browser, is one example. Other examples are the Linux operating system, and Apache web server software.

The R Project for Statistical Computing

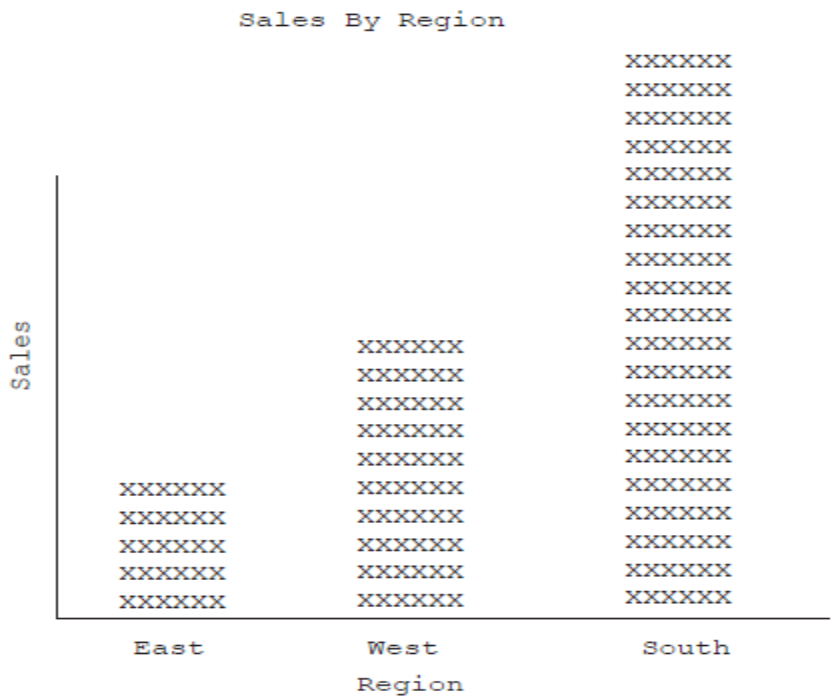
Open-source has come to the world of advanced analytics in the form of the R Project for Statistical Computing, also known simply as R. **R is a free, open-source analytics package that competes directly with commercial analytic tools**. R is a descendent from the original "S." S was an early language for statistical analysis that was developed decades ago. The name R appears to be derived both due to the software being an update to S and also because the original authors names (**Robert Gentleman and Ross Ihaka**) began with R.

Features of R:

- R has stronger **object-oriented programming facilities** than most statistical computing languages.
- R is **easily extensible through functions and extensions** and can link with common programming platforms like C++ and Java, which makes it possible to embed R within applications.
- Most **commercial analytic tools have enabled R** to be executed within their toolsets.
- Major advantage of R is **its extensibility**. Developers can easily write their own software and distribute it in the form of add-on packages. Because of the relative ease of creating these packages, thousands of R packages exist.

The History of Data Visualization

In the world of analytics, **visualization refers to charts, graphs, and tables that display data.** Until the age of computers, graphs were drawn by hand. Computers revolutionized the methods and ease of creating visualizations. Today there are visualization tools that allow interaction with graphics to explore and analyze data in new and powerful ways.



Modern Visualization Tools

Visualization tools enable the analytic professional or business user to develop a new story in an interactive, visual paradigm. Today's visualization tools allow multiple tabs of graphs and charts to be linked to the underlying data. Even more important, the tabs, graphs, and charts can be linked to each other. If a user clicks on the bar for the Northeast region, all the other graphs will instantaneously adjust and show only Northeast data as well.

Analysis Versus Reporting

Too many organizations mistakenly equate reporting with analysis.

Reports **are important and can be valuable. Reports used correctly will add value**. But reports have their limits, and it is important to understand what they are. In the end, an organization will need both reporting and analysis to succeed in taming big data, just as both reporting and analysis have been utilized to tame every other data source that's come along in the past. The key is to understand the

difference between a report and an analysis. It is also critical to understand how they both fit together.

Reporting

A reporting environment is also called **business intelligence (BI) environment**. Such an environment is where users go to select the reports they want to run, get the reports executed, and view the results. The reports may contain tables, graphs, and charts in any combination. The key factors that define a report include:

- A report will provide back to the user the data that was asked for.
- That data will be provided in a standardized, predefined format.
- There is no person involved in generating a report outside of the user who requested the report through his or her reporting interface.
- As a result, reports are fairly inflexible.

Complicated report templates can be created with a variety of prompts and filters. There may be many options in such reports, but within the constraints of those predefined options they are fairly inflexible. The average user is not typically able to generate a completely new report or fix how the predefined prompts and filters work. The user can simply fill in the prompts and filters that are already in place.

Analysis

The key points that define an analysis are:

- An analysis provides answers to the questions being asked.
- An analysis process takes any steps needed to get the answers to those questions.
- An analysis is therefore customized to the specific questions being addressed.
- An analysis involves a person who guides the process.

- By its very nature, the analysis process is flexible.

An analysis is really about saying: "I've heard the problem. I'm going to put together what is needed to address the problem." It's an interactive process of a person tackling a problem, finding the data required to get an answer, analyzing that data, and interpreting the results in order to provide a recommendation for action.

The differences between analysis and reporting are summarized in Table:

Reporting . . .	Analysis . . .
Provides data	Provides answers
Provides what is asked for	Provides what is needed
Is typically standardized	Is typically customized
Does not involve a person	Involves a person
Is fairly inflexible	Is extremely flexible

Interplay between reporting and analysis is common and necessary. In fact, each makes the other more effective.

For example, consider a sales manager with a basic sales summary report showing monthly sales by region. It is a very simple report that he looks at every day so he can get a feel for whether or not the business is on track. One day he sees something incredibly unusual that he doesn't understand. As a result, he walks down the hall and alerts the analytics team that there's something unusual on the sales summary report. He asks them to find out what's going on. His request based on that report just generate an analysis, which is exactly what it should be doing.

On the flip side, consider the analytic professional assigned to investigate that problem. She goes through and identifies what some of the underlying causes are. She comes back and shows the sales manager what she found. The manager may comment that the data she just put together is very, very useful. Her analysis of a problem today has led to a new standard report. She automates what she did, and it becomes a standard report moving forward.

Modern Data Analytic Tools

1. R Programming

- R is the leading analytics tool in the industry and widely used for statistics and data modeling.
- It can easily manipulate data and present in different ways. It has exceeded SAS in many ways like capacity of data, performance and outcome.
- R compiles and runs on a wide variety of platforms like -UNIX, Windows and MacOS.
- It has 11,556 packages and allows to browse the packages by categories.
- R also provides tools to automatically install all packages as per user requirement, which can also be well assembled with Big data.

2. Tableau Public:

- Tableau Public is a free software that connects any data source to corporate Data Warehouse, Microsoft Excel or web-based data, and creates data visualizations, maps, dashboards etc. with real-time updates presenting on web.
- They can also be shared through social media or with the client. It allows the access to download the file in different formats.

- Tableau's Big Data capabilities makes them important and one can analyze and visualize data better than any other data visualization software in the market.

3.Python

- Python is an object-oriented scripting language which is easy to read, write, maintain and is a free open source tool.
- It was developed by Guido van Rossum in late 1980's which supports both functional and structured programming methods.
- Python is easy to learn as it is very similar to JavaScript, Ruby, and PHP. Also, Python has very good machine learning libraries.
- Another important feature of Python is that it can be assembled on any platform like SQL server, a MongoDB database or JSON.
- Python can also handle text data very well.

4. SAS

- SAS is a programming environment and language for data manipulation and a leader in analytics, developed by the SAS Institute in 1966 and further developed in 1980's and 1990's.
- SAS is easily accessible, manageable and can analyze data from any sources.
- SAS introduced a large set of products in 2011 for customer intelligence and numerous SAS modules for web, social media and marketing analytics that is widely used for profiling customers and prospects.
- It can also predict their behaviors, manage, and optimize communications.

5. Apache Spark

- The University of California, Berkeley's AMP Lab, developed Apache in 2009. Apache Spark is a fast large-scale data processing engine and executes applications in Hadoop clusters 100 times faster.
- Spark is built on data science and its concept makes data science effortless.
- It is also popular for data pipelines and machine learning models development.
- Spark also includes a library – MLlib, that provides a progressive set of machine learning algorithms for repetitive data science techniques like Classification, Regression, Collaborative Filtering, Clustering, etc.